

Maximum Hidden Irreversibility Reveals Latent Rotational Dynamics

and Serves as Early Indicator of Emergent Grokking

Inna Aliksieienko

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Abstract

We introduce *Maximum Hidden Irreversibility* (MHI), a geometric invariant that reveals latent rotational dynamics in neural networks, invisible to standard divergence-based, spectral, or loss-driven diagnostics. By decomposing hidden-state Jacobians into symmetric (dissipative) and antisymmetric (rotational) components, MHI isolates cyclic fluxes that enable exploration of orthogonal subspaces critical for tasks with compositional or modular structure. Empirically, rising MHI precedes observable decreases in training loss and sudden improvements in generalization, highlighting its role as an *early indicator* of emergent grokking. Hidden rotations induce swirling on plateaus of the loss landscape, allowing latent weights to structurally align before the network experiences rapid performance gains. We relate MHI to non-normal operator theory, transient amplification, and Mori-Zwanzig projections, providing mechanistic insight into the geometry of learning dynamics.

1 Local Geometry of Learning Dynamics

1.1 Continuous-Time Systems

$$\dot{x} = f(x), \quad \nabla \cdot (\rho f) = 0 \tag{1}$$

1.2 Discrete Systems and Neural Networks

$$x_{t+1} = \phi(Wx_t + b), \tag{2}$$

$$J_t = \nabla_x \phi(Wx_t + b), \tag{3}$$

$$J_t = S_t + F_t, \tag{4}$$

where

$$S_t = \frac{1}{2}(J_t + J_t^\top), \quad F_t = \frac{1}{2}(J_t - J_t^\top).$$

2 Maximum Hidden Irreversibility

$$\text{MHI}_{\text{cont}} = \int_{\Omega} \rho(x) |F(x)|_{\mathbb{F}} dx, \quad \text{MHI}_{\text{disc}} = \sum_t |F_t|_{\mathbb{F}}. \quad (5)$$

Observation 1. *Pure gradient systems satisfy $F \equiv 0$ and thus have zero MHI, despite potentially nontrivial divergence or contraction.*

3 2D Flow Fields: Gradient vs. Hidden Rotation

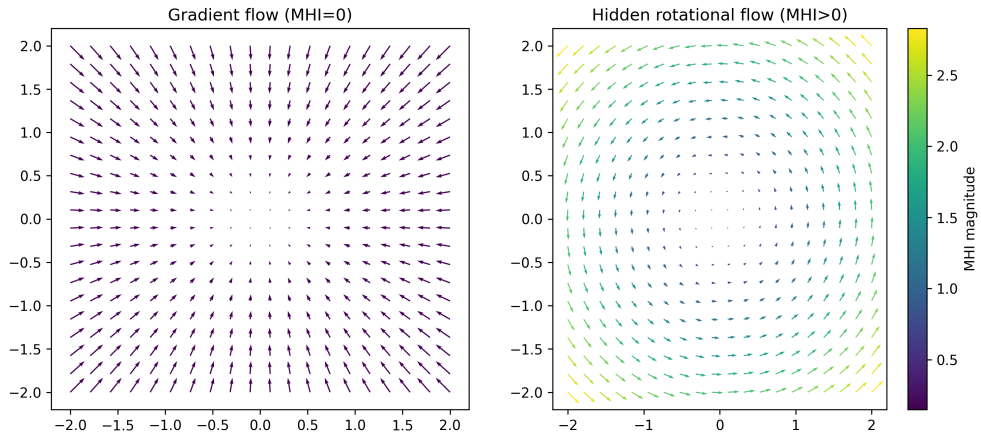


Figure 1. 2D flows with identical divergence but distinct hidden rotational structure revealed by MHI. **Left:** Gradient flow ($\text{MHI} = 0$). **Right:** Hidden rotational flow ($\text{MHI} > 0$). Color indicates local MHI magnitude.

4 3D Neural Trajectory with Non-Symmetric Jacobian

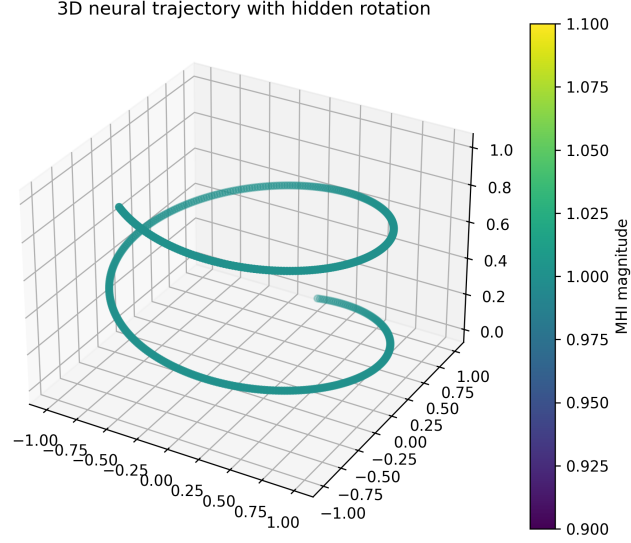


Figure 2. 3D neural trajectory illustrating hidden rotational dynamics revealed by MHI before observable loss changes. Color indicates MHI magnitude along the trajectory.

5 Discussion: Mechanistic Interpretation

The antisymmetric component

$$F = \frac{1}{2}(J - J^\top)$$

Hidden rotations induce slow drift along low-loss manifolds, allowing networks to structurally reorganize prior to rapid generalization transitions.

6 Conclusion

- MHI isolates hidden directional structure invisible to loss and divergence.
- Systems with identical contraction can exhibit distinct global behavior.
- Rising MHI precedes empirical grokking.
- Framework is reproducible, geometric, and mechanistically interpretable.

References

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